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Reg. No. | | | | | | | |

Time : 3.00 hrs.

Quarterly Examination - 2025

CHEMISTRY

Max. Marks : 70

PART - I

I. (i) Answer all questions. **15x1=15**

(ii) Choose the best answer from given four alternatives and write the option code and the corresponding answer.

1. Elements like silicon and germanium to be used as a semi-conductor is purified by
   a) heating under vacuum b) Van Arkel method c) Zone refining d) Electrolytic refining
2. Match items in column-I with items in column-II and assign the correct code.

Column-I		Column-II
A) Borazole	-	1. $B(OH)_3$
B) Boric Acid	-	2. $B_3N_3H_6$
C) Quartz	-	3. $Na_2[B_4O_3(OH)_4]\ H_2O$
D) Borax	-	4. SiO_2

	A	B	C	D
a)	2	1	4	3
b)	1	2	4	3
c)	1	2	3	4
d)	none of these			

3. Most easily liquefiable gas a) Ar b) He c) Ne d) Kr
4. Which of the following lanthanoid ion is diamagnetic. a) Eu^{2+} b) Ce^{2+} c) Yb^{2+} d) Sm^{2+}
5. CSCl has bcc arrangement, its unit cell edge length is 400pm, its interionic distance is
   a) 800pm b) $\left(\frac{\sqrt{3}}{2}\right)$ x 400pm c) 400pm d) $\sqrt{3}$ x 100pm
6. For the reaction $2A+B \rightarrow 3C+D$ which of the following does not express the reaction rate.
   a) $\frac{d[D]}{dt}$ b) $-\frac{d[A]}{2dt}$ c) $-\frac{d[c]}{3dt}$ d) $\frac{d[B]}{dt}$
7. Which of the following fluro-compounds most likely to behave as a Lewis base?
   a) BF_3 b) CF_4 c) SiF_4 d) PF_3
8. Which of following used in medicine in Hypnotic historically known as "Hypnone"?
   a) Acetophenone b) formaldehyde c) paraldehyde d) Benzaldehyde
9. On reacting with neutral $FeCl_3$, phenol gives
   a) red colour b) dark green colour c) violet colour d)no colouration
10. Which of the following compound is used as antifreeze in automobile?
    a) Ethanol b) Ethylene glycol c) Methanol d) Neopentyl alcohol
11. Assertion: Bond dissociation energy of fluorine is greater than chlorine gas.
    Reason: Chlorine has more electronic repulsion than fluorine.
    a) Both assertion and reason are true and reason is the correct explanation of assertion b) Both assertion and reason are true and reason is not correct explanation of assertion
    c) Assertion is true but reason is false d) Both assertion and reason are false.
12. Which of the metal is extracted by Hall-Herrold process?
    a) Ni b) Zn c) Al d) Cu

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13. In Wolf Kishner reduction _____ acts as reducing agent and _____ acts as catalyst.
a) Hydrazine & Sodium ethoxide b) Ketone & Hydrazine c) Sodium ethoxide & Hydrazine
d) Aldehyde & Hydrazine

14. In given transition metal ion series which series has all the metal ion in $3d^2$ electronic configuration. (Atomic number: Ti = 22, V = 23, Cr = 24, Mn = 25)
a) Ti^{2+}, V^{3+}, Cr4+, Mn^{6+} b) Ti^{3+}, V^{2+}, Cr^{3+}, Mn^{4+} c) Ti^{+}, V^{4+}, Cr^{0+}, Mn^{7+} d) Ti^{4+}, V^{4+}, Cr^{2+}, Mn^{3+}

15. The conjugate acid of NH_2^- is a) NH^{2-} b) NH_2 c) NH_3 d) NH_4^+

PART - II

Note: Answer any six questions. Question No.24 is compulsory. **6x2=12**

16. Name the methods to extract following pure metals. (i) Zirconium (ii) Nickel
17. Co is a reducing agent. Justify with example.
18. What is the hybridization of iodine in IF_7 and give its structure?
19. Which is more stable Fe^{2+} & Fe^{3+}? Explain.
20. Calculate the number of atoms in FCC unit cell.
21. Give two example for zero order reaction.
22. Write the expression for solubility product of Hg_2Cl_2.
23. Explain Williamson Synthesis.
24. Which compounds give 2-methyl propane on Clemmenson reduction?

PART - III

Note: Answer any 6 questions. Question No.33 is compulsory. **6x3=18**

25. Write the uses of Boric acid.
26. Differentiate between Order and Molecularity.
27. Why d-block elements form complexes?
28. Define common ion effect with example.
29. Write a note on Fischer-Tropsch Synthesis.
30. What are the limitations of Ellingham diagram?
31. How will you prepare acetaldehyde from acetyl chloride? Write equation.
32. Write short note on Lucas test.
33. In a first order reaction Aproduct 60% of A decomposes in 40 min. What is half-life period?

PART - IV

Note: Answer all questions. **5x5=25**

34. a) Write short note on the following with example:
1. (i) Gangue (ii) Slag (2) 2. Explain froth-floatation process with diagram. (3) **(OR)**
b) Differentiate Lanthanoids and Actinoids. (5)

35. a)(i) How will you identify Borate radical (3)
(ii) Give reason to support sulphuric acid is dehydrating agent. (2) **(OR)**
b) (i) Derive the expression for Oswald dilution law. (3) (ii) Define unit cell. (2)

36. a) Derive expression for zero order integrated rate constant equation. A product. **(OR)**
b) (i) Write short note on Frenkel's defect. (3) (ii) Write Bragg's equation and write its terms. (2)

37. a)(i) What is urotropine? How it is prepared? (3)
(ii) Formic Acid reduces Tollen's reagent. Give reasons. (2) **(OR)**
b) Explain the mechanism of Aldol condensation.

38. a)(i) How will you convert Boric acid into Boron Nitride? (2) (ii) Give the uses of Helium. (3) **(OR)**
b) An organic compound (A) of molecular formula C_6H_6 on reaction with propylene. $AlCl_3$ gives (B) in presence of H_3PO_4 at 523k. (B) on air oxidation gives (C) of molecular formula $C_9H_{12}O_2$. (C) on acidification with H_2SO_4 gives (D). Identify A, B, C, D and write equations.

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